

Ayush Saha

[Github](#) [LinkedIn](#) [Portfolio](#)

Email: ayushsaha184@gmail.com

Mobile: +91-7679081423

SUMMARY

AI Engineer specializing in **Agentic RAG** and **Model Context Protocol (MCP)** server development. Experienced architecting production-grade **multi-agent pipelines**, **RESTful APIs**, and **LLM-powered backends** using FastAPI, LangChain, and LangGraph. Focused on building reliable, cost-efficient, and observable AI systems with measurable improvements in latency, accuracy, and cost.

EDUCATION

Siliguri Institute of Technology

Bachelor of Engineering – Computer Science and Engineering

West Bengal

2023 – 2027

TECHNICAL SKILLS

Languages: Go, Python, Java

Frameworks/Libraries: FastAPI, LangChain, LangGraph, LlamaIndex, RESTful API Design

Databases: PostgreSQL, MySQL, Redis, Qdrant (Vector Search)

Cloud/DevOps: Docker, Vercel, Supabase, Neon, Render, Git, CI/CD

Generative AI: Retrieval-Augmented Generation (RAG), Agentic AI Systems, Model Context Protocol (MCP), Prompt Engineering, NLP, Transformers, Open-Source LLMs, LLM Evaluation

Coursework: Operating Systems, Database Management Systems, Computer Networks, Object-Oriented Programming

PROJECTS

PR Review Agent | Python, LangGraph, FastAPI, GitHub REST API, Redis

[Github](#)

- Architected a **multi-agent code review system** using LangGraph that orchestrates four specialized AI agents (security, code quality, tests, documentation) to automate GitHub pull request reviews, cutting manual review overhead by **60%**.
- Designed a **human-in-the-loop approval policy** with confidence- and severity-based gating that routes critical, uncertain, or incomplete findings to human review, preventing **100%** of high-risk auto-posted comments while preserving review velocity.
- Built **repository-grounded retrieval** combining hybrid vector and full-text search to ground agent responses in real code context, improving review relevance and reducing false-positive findings by **40%**.
- Engineered a secure, idempotent **webhook pipeline** with **HMAC-SHA256** signature verification and Redis/ARQ-based asynchronous job queuing to process GitHub events reliably under load, eliminating duplicate reviews.

LLM Evaluation & Accuracy MCP | Go, Model Context Protocol (MCP), FastAPI, Redis

[Github](#)

- Built a dependency-free, **single-binary MCP server** in Go exposing **16 native evaluation tools** for prompt regression testing and RAG pipeline scoring, reducing deployment complexity to a single executable.
- Implemented **pure-Go heuristic scoring metrics** (**Exact Match**, **Levenshtein Distance**, **Token-Overlap**, **Hallucination Detection**, **Context Precision/Recall**, **Lexical Toxicity**) to evaluate RAG outputs **without external LLM calls**, cutting per-evaluation latency by **80%** and **API costs to zero** for standard runs.
- Designed an **async API-worker architecture** with Redis-backed job queuing and configurable polling, enabling **concurrent, high-throughput batch evaluation** without blocking the MCP client.

Agentic RAG Doc Assistant | Python, LangChain, LangGraph, Qdrant, FastAPI, Redis, Supabase [Demo](#) | [Github](#)

- Engineered a production-grade **Agentic RAG** platform handling **100K+ documents** at **60–120 queries/min** throughput using Python, FastAPI, and Supabase.
- Designed a **5-stage multi-agent retrieval pipeline** with automated **query classification** and timeout guardrails, improving **answer reliability** by **~35%** and **citation precision** by **~25%**.
- Optimized **document ingestion** to **60+ docs/sec** (6–8x faster) via threaded loading, process-based **semantic chunking**, and batched embedding generation.
- Reduced **LLM API costs** by **50%** and **median response latency** by **55%** with a **four-level caching architecture** spanning KV, retrieval, LLM response, and human-validated output caches.